

Introduction to Clustering Complex Data

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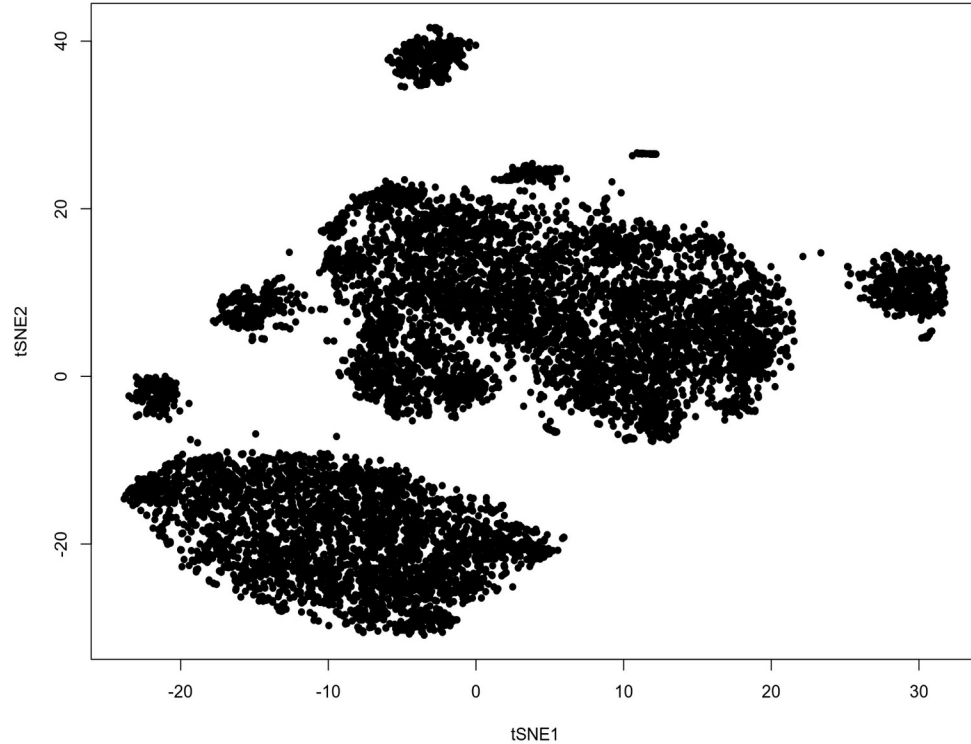
@osolkov.bsky.social



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<https://nikolay-osolkov.com>

Image adapted from www.stratio.com

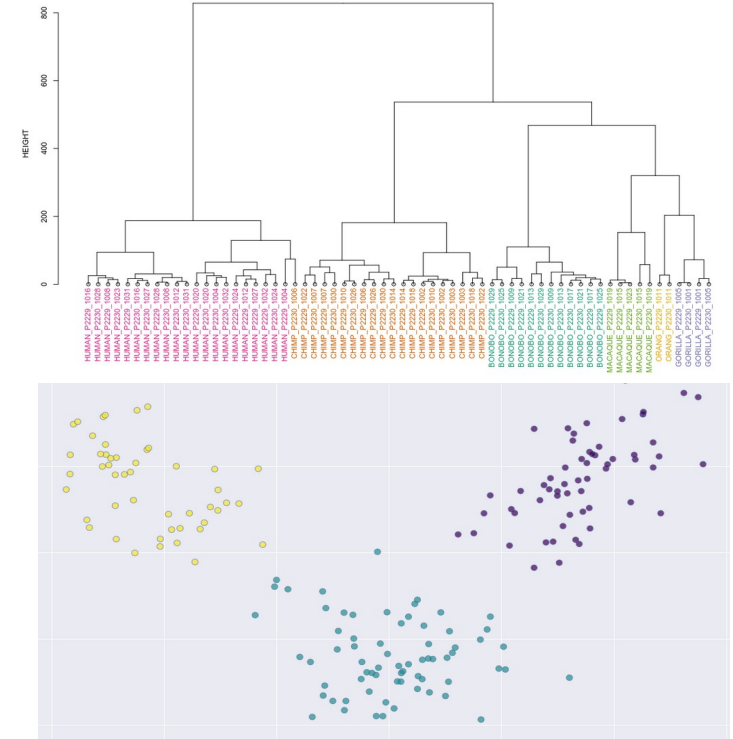
Exploratory data analysis (EDA): dimensionality reduction and clustering



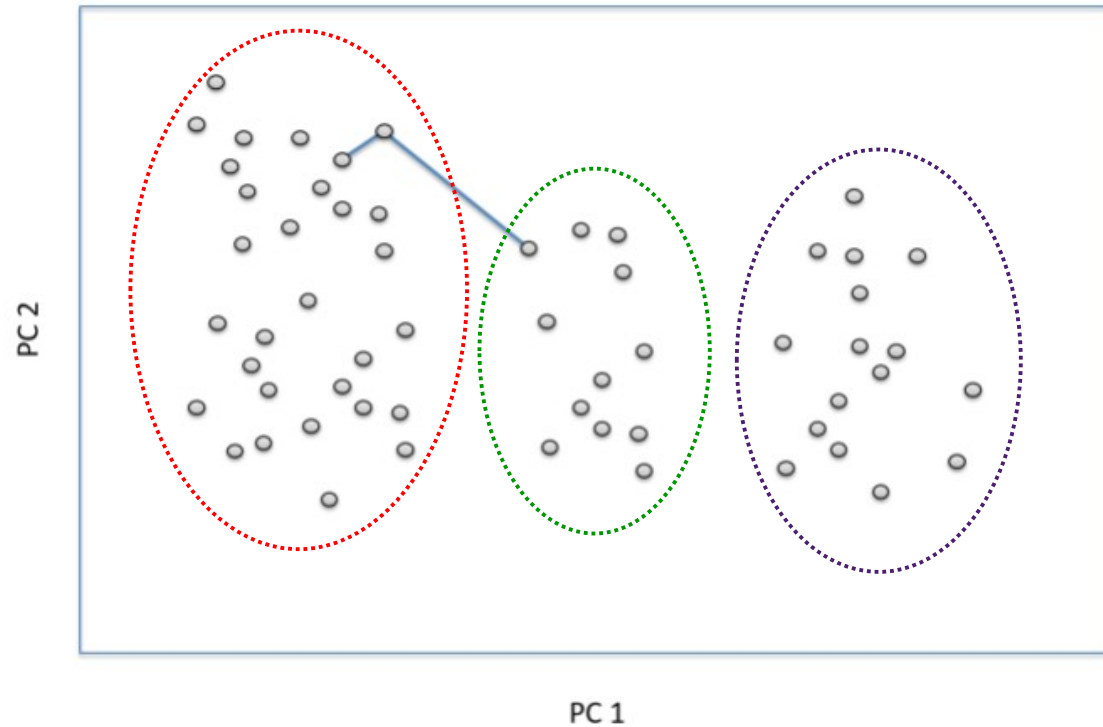
Dimension reduction:
data pre-processing + visualization



**Working
hand in hand**



Cluster analysis:
formal grouping of data points
based on their similarity or difference



Connect objects in some way based on similarity

Metrics intended for real-valued vector spaces:

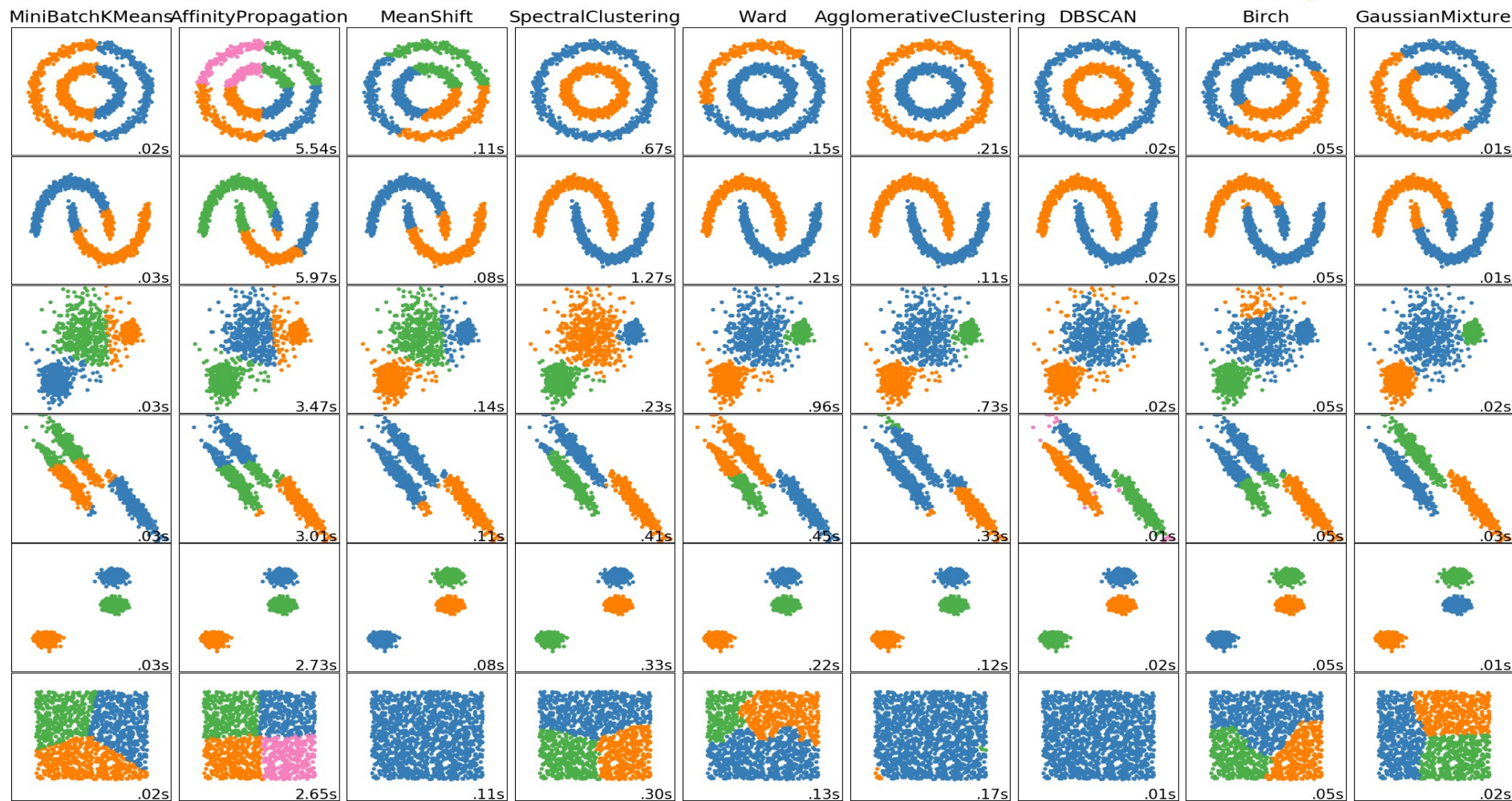
identifier	class name	args	distance function
"euclidean"	EuclideanDistance	•	$\sqrt{\sum (x - y)^2}$
"manhattan"	ManhattanDistance	•	$\sum x - y $
"chebyshev"	ChebyshevDistance	•	$\max(x - y)$
"minkowski"	MinkowskiDistance	p, w	$\sum (w * x - y ^p)^{1/p}$
"seuclidean"	SEuclideanDistance	V	$\sqrt{\sum (x - y)^2 / V}$
"mahalanobis"	MahalanobisDistance	V or VI	$\sqrt{(x - y)' V^{-1} (x - y)}$

Metrics intended for two-dimensional vector spaces: Note that the haversine distance metric requires data in the form of [latitude, longitude] and both inputs and outputs are in units of radians.

identifier	class name	distance function
"haversine"	HaversineDistance	$2 \arcsin(\sqrt{\sin^2(0.5 * dx) + \cos(x1) \cos(x2) \sin^2(0.5 * dy)})$

Metrics intended for integer-valued vector spaces: Though intended for integer-valued vectors, these are also valid metrics in the case of real-valued vectors.

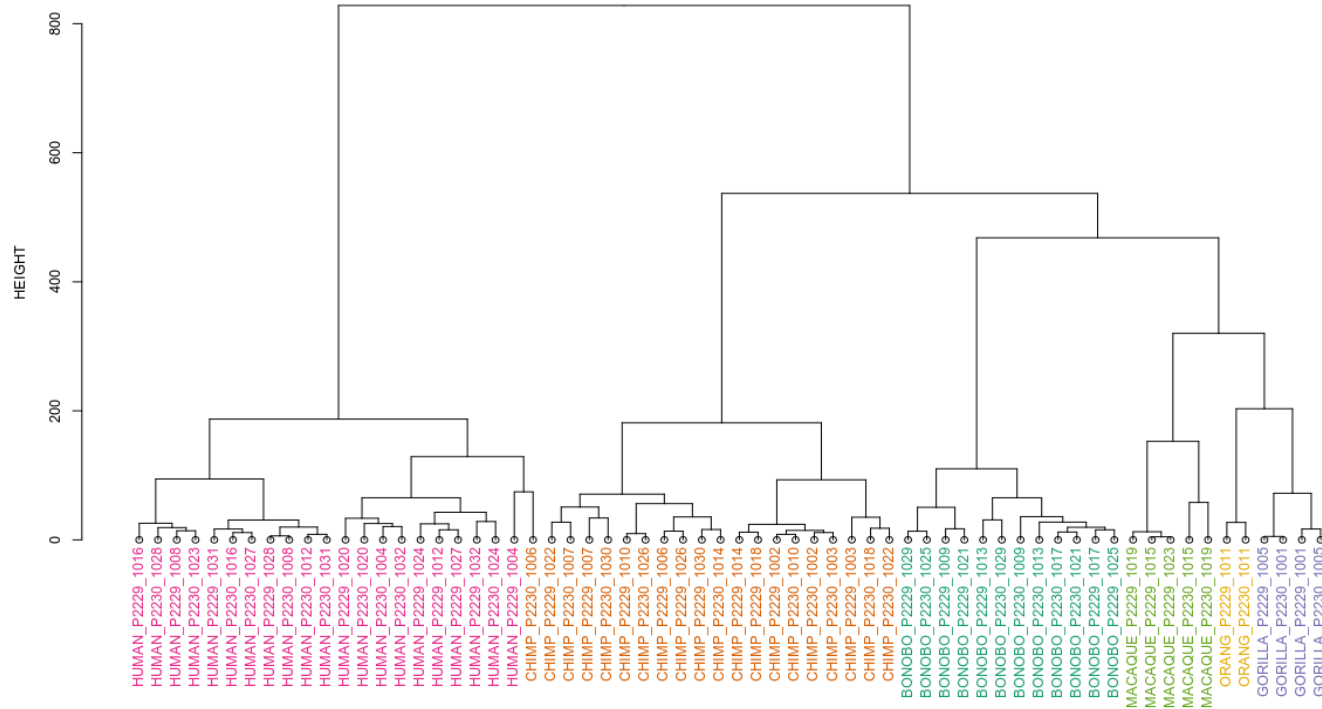
identifier	class name	distance function
"hamming"	HammingDistance	$N_{\text{unequal}}(x, y) / N_{\text{tot}}$
"canberra"	CanberraDistance	$\sum x - y / (x + y)$
"braycurtis"	BrayCurtisDistance	$\sum x - y / (\sum x + \sum y)$



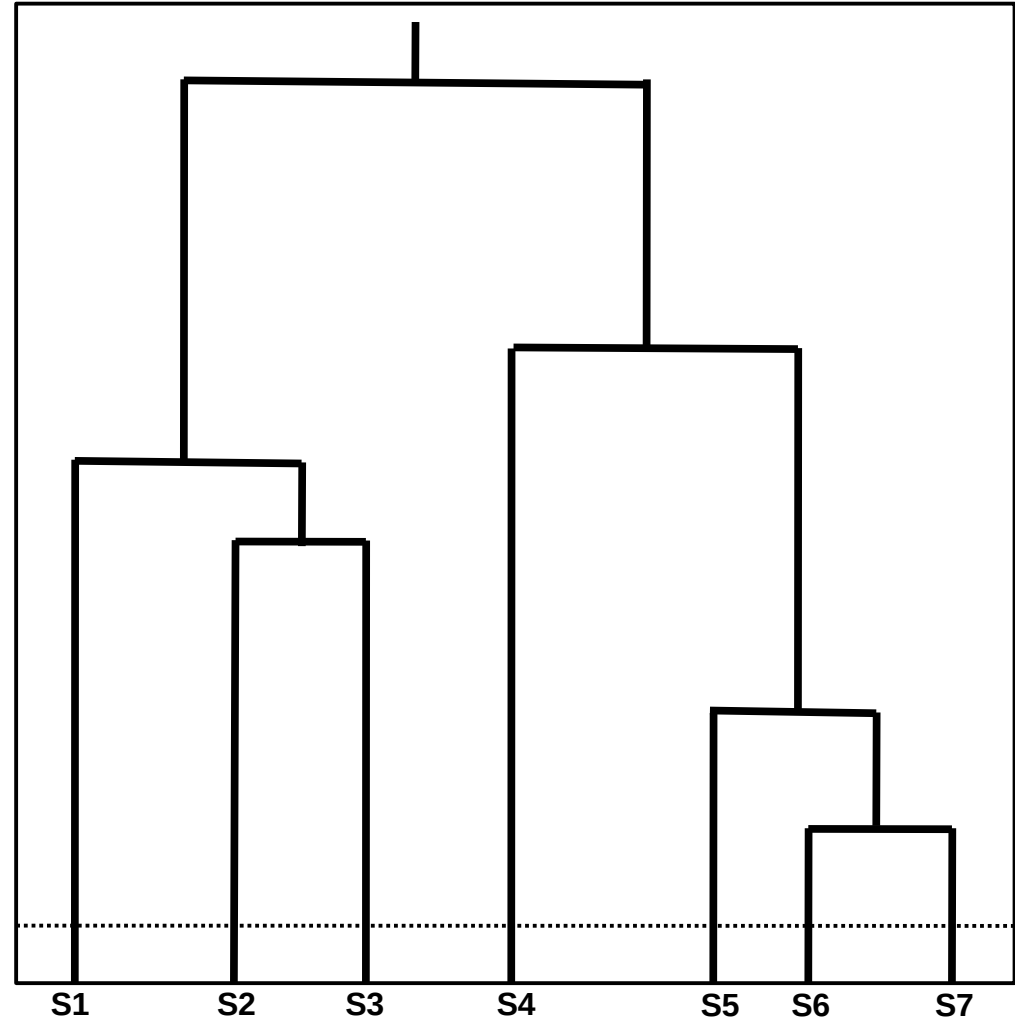
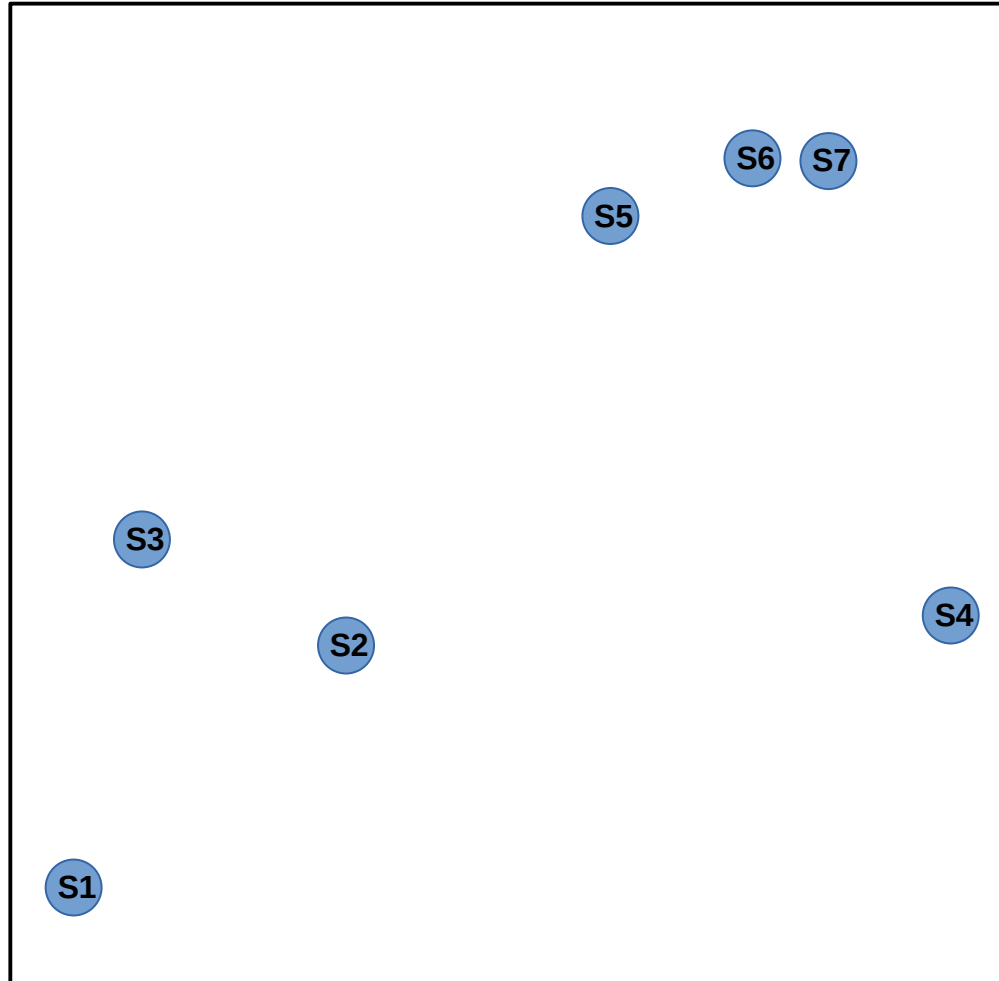
1) Hierarchical Clustering, 2) Partitioning Clustering, 3) Density-Based Clustering, 4) Graph-Based Clustering

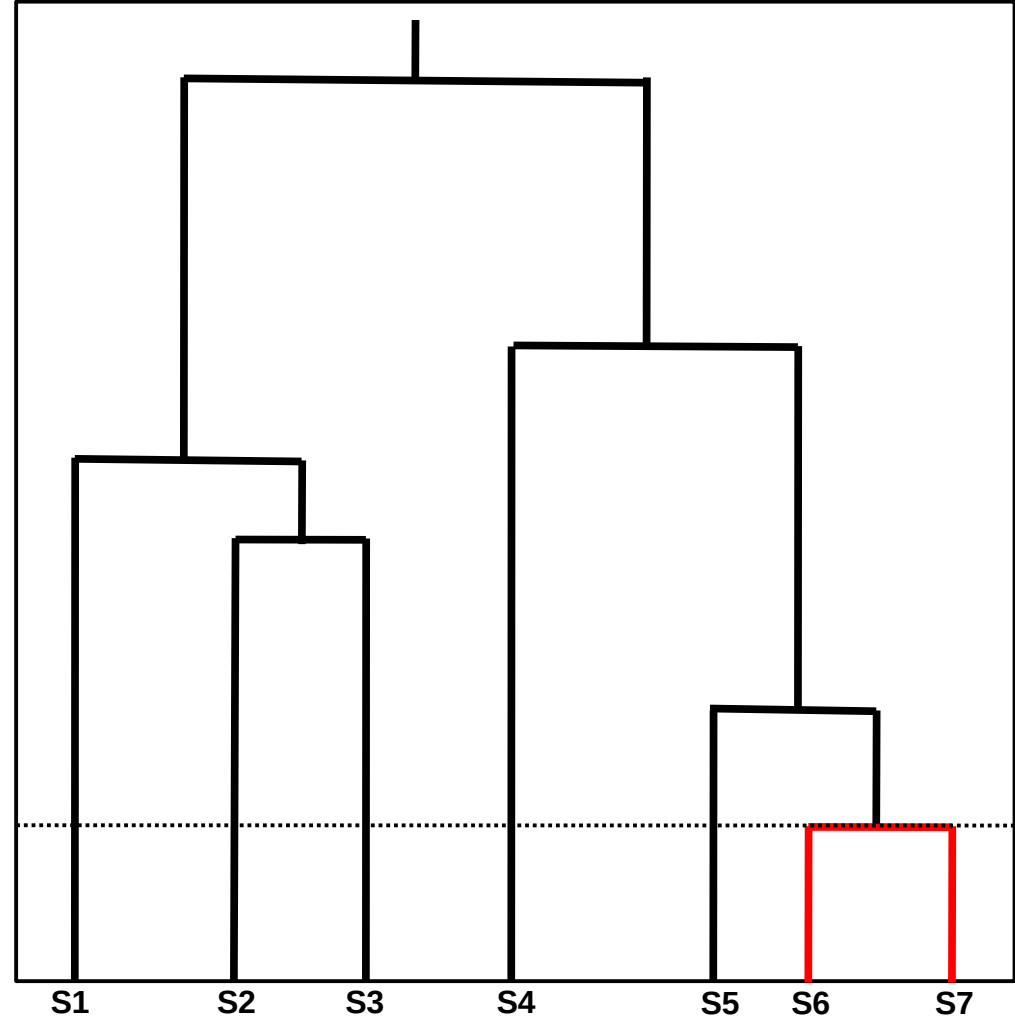
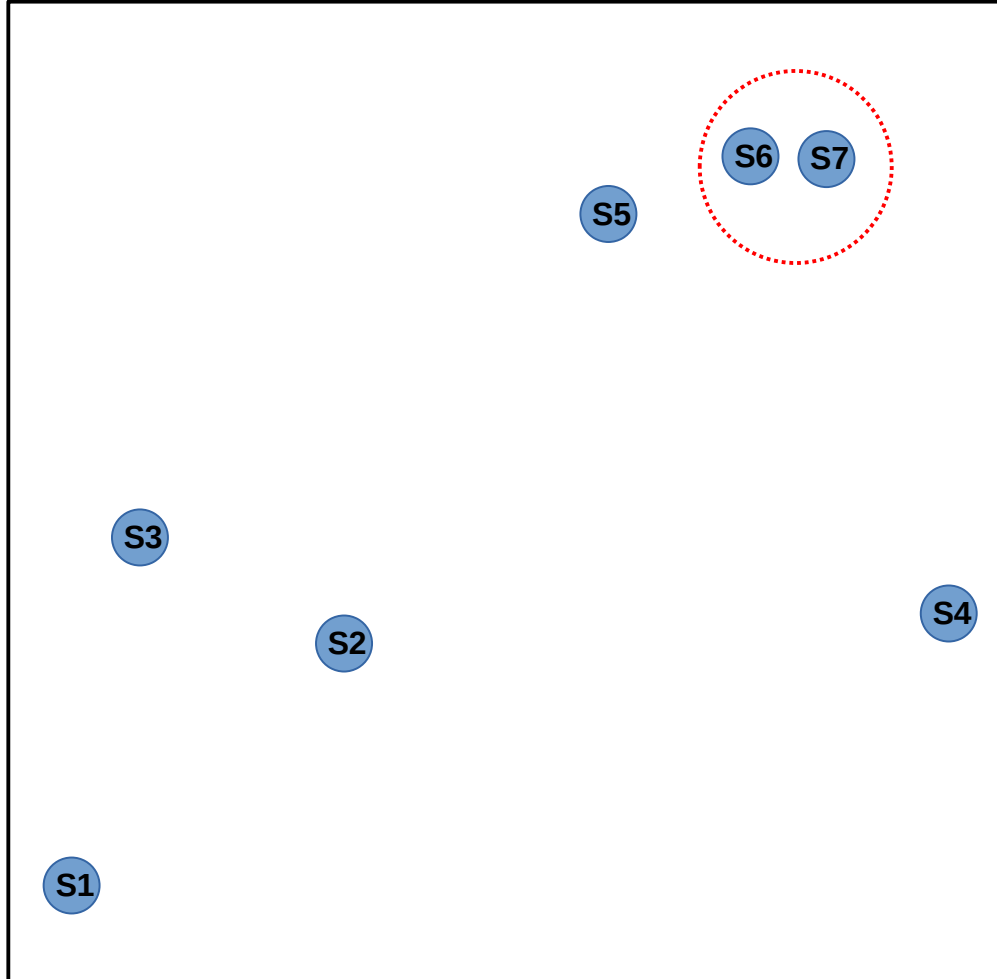
Hierarchical clustering

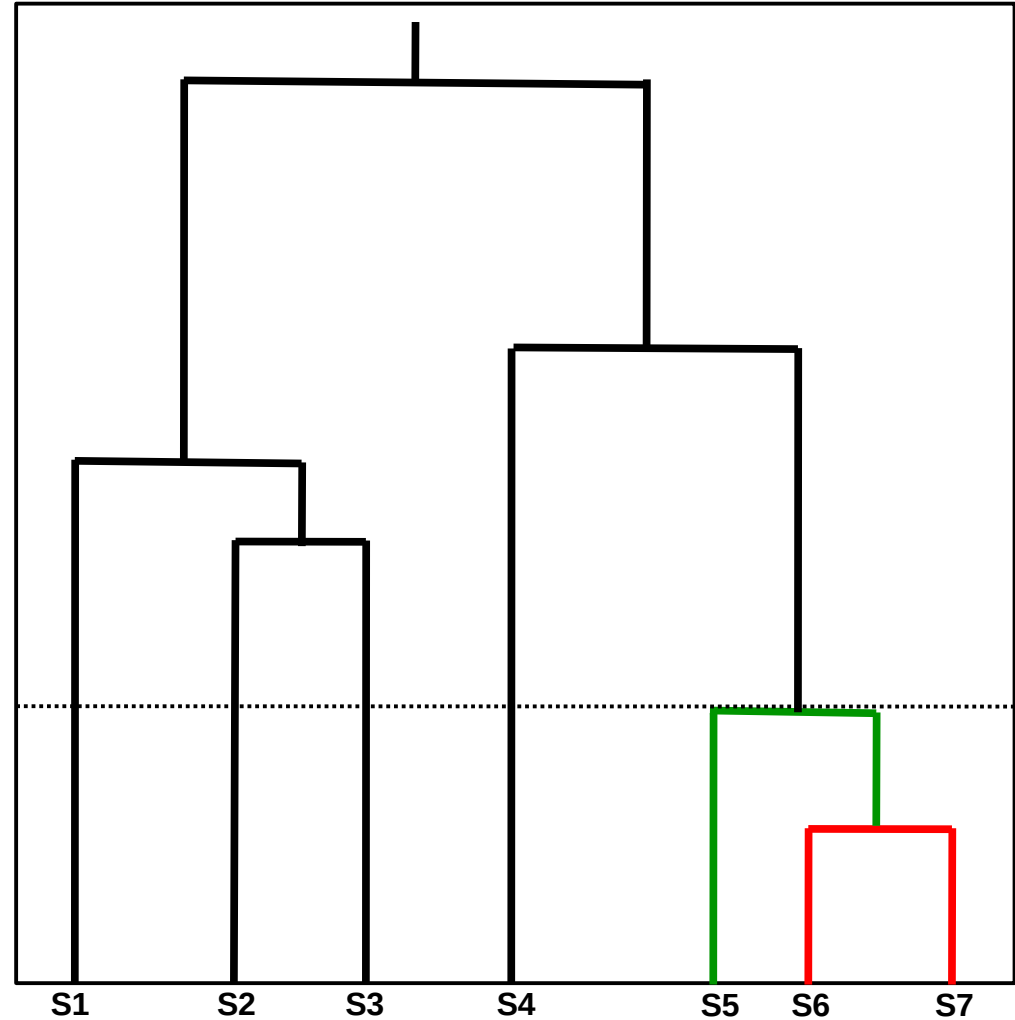
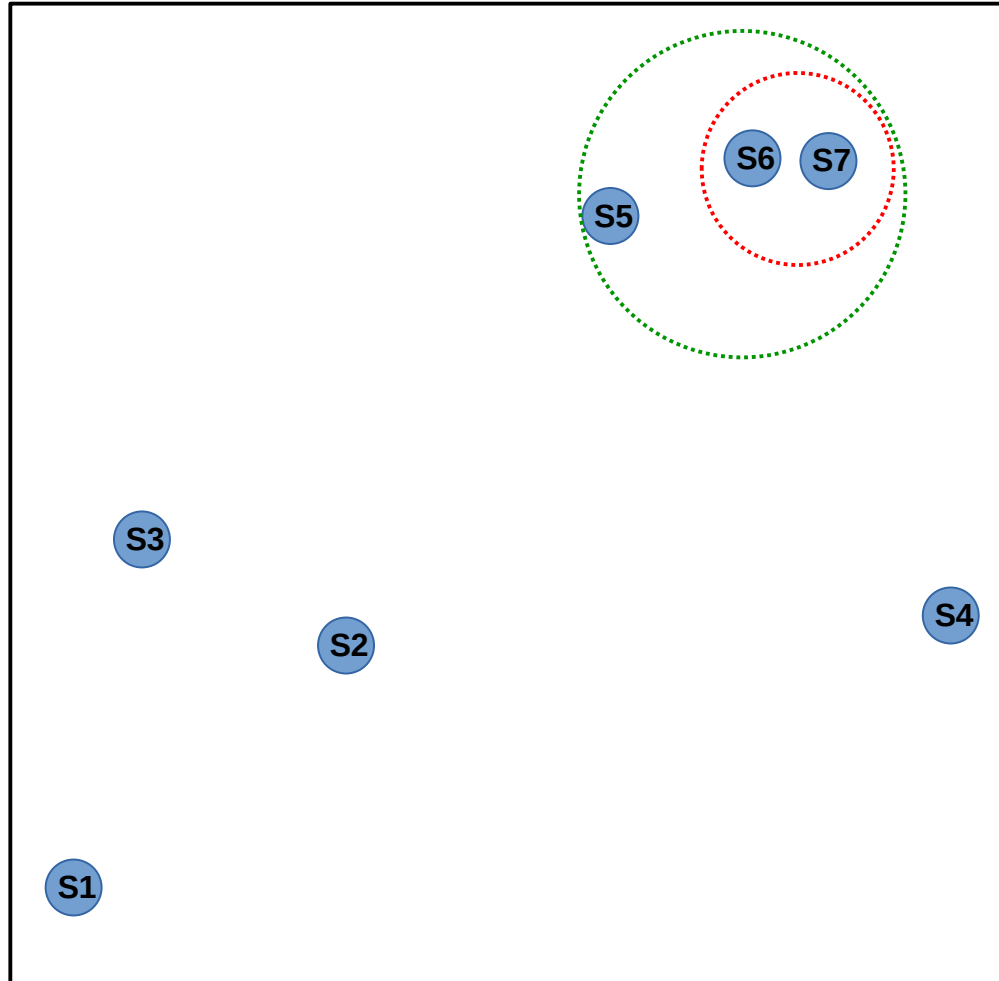
Clusters are nested (higher-ranked clusters contain lower-ranked clusters),
in other words tree-like structure (dendrogram)

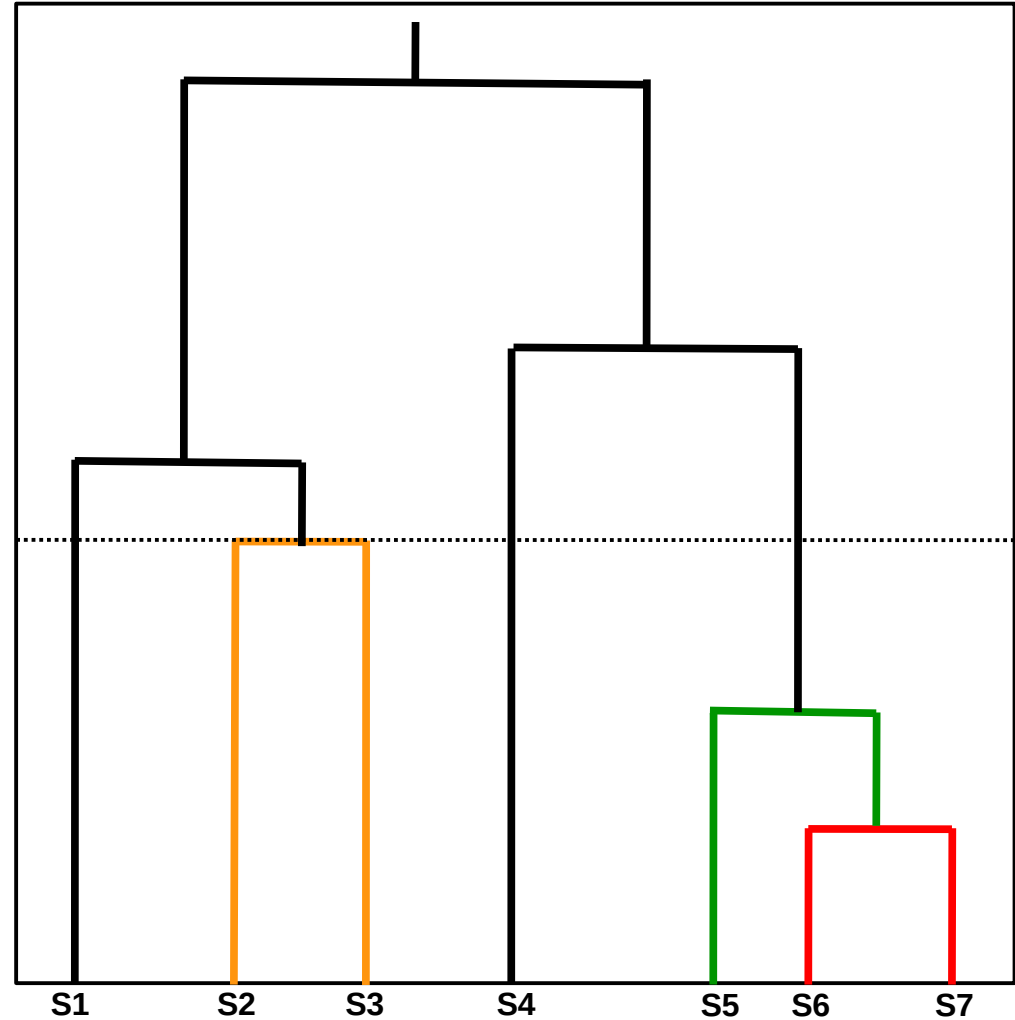
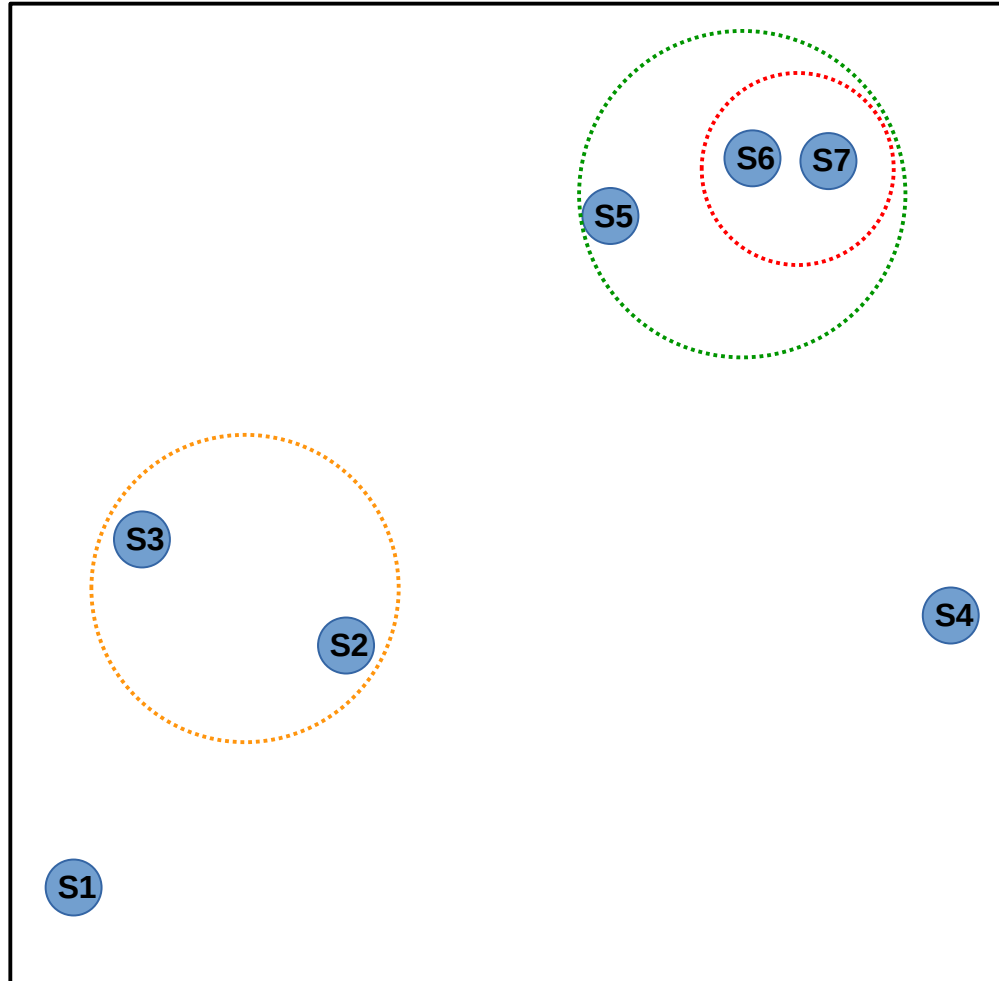


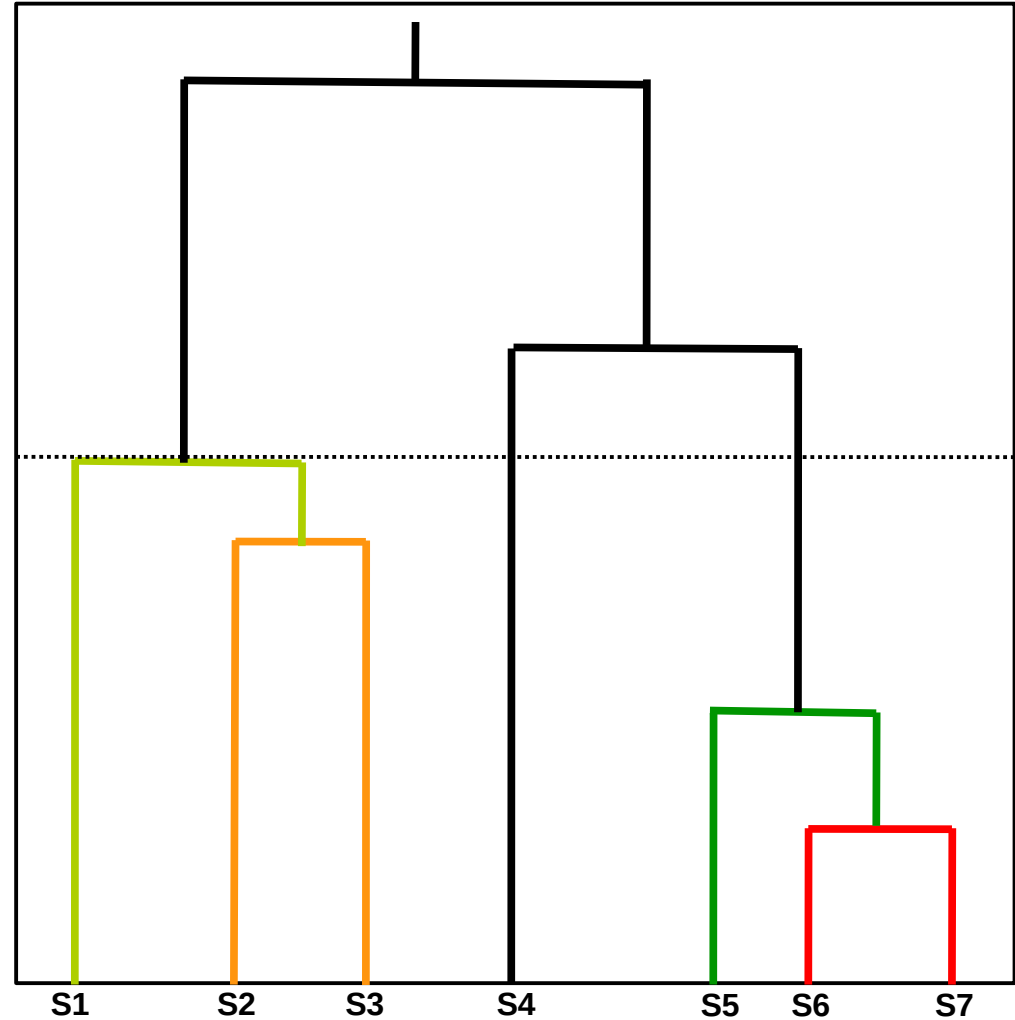
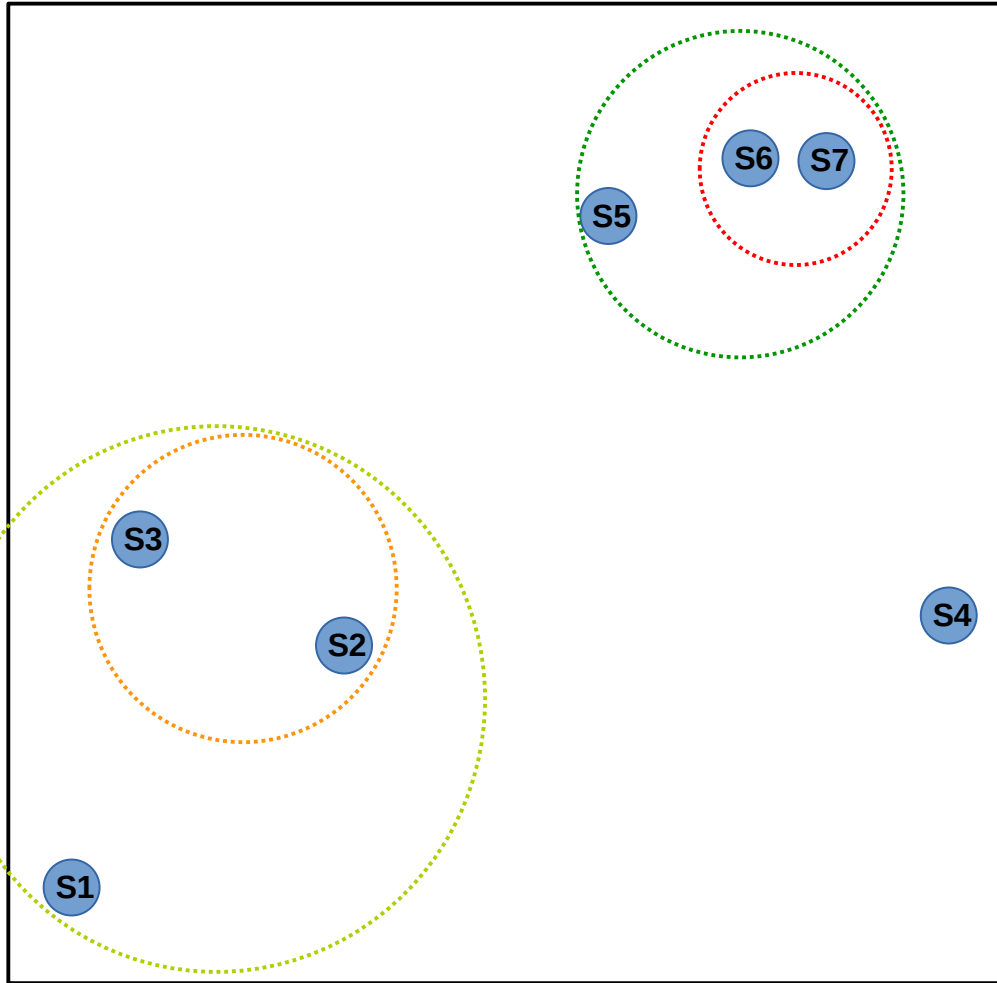
There are agglomerative (bottom-up approach) and divisive (top-down approach)
types of hierarchical clustering

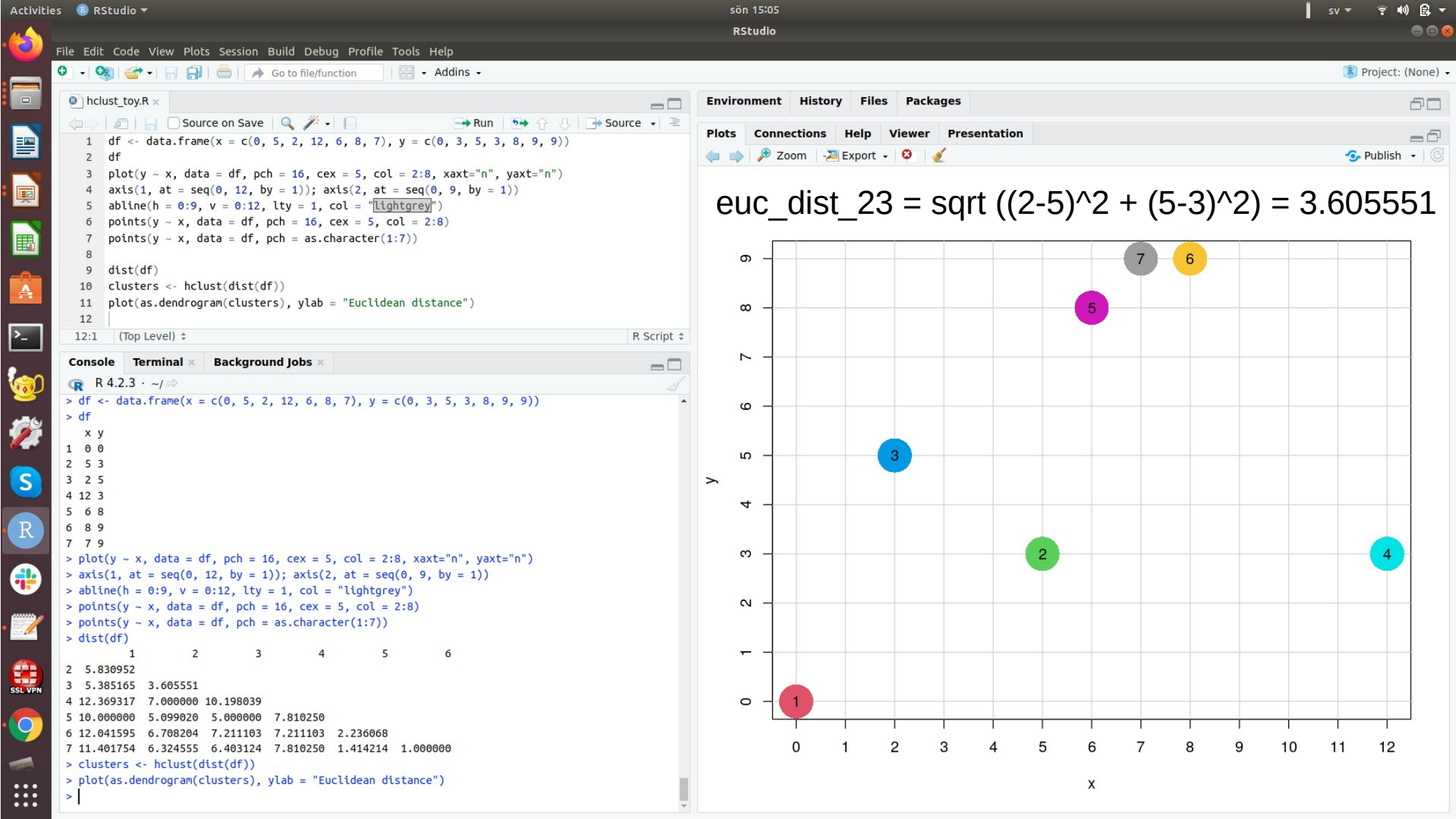


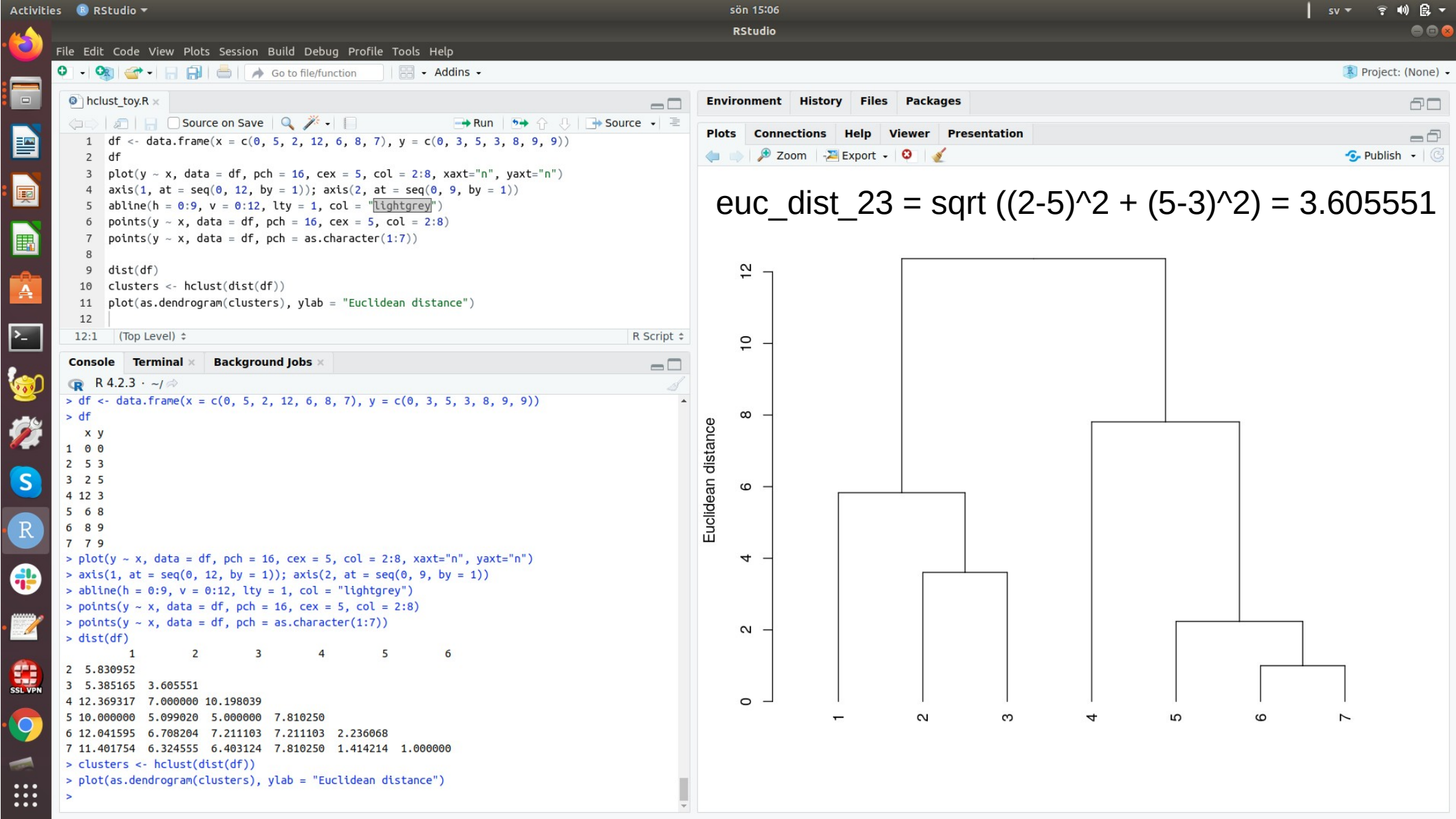












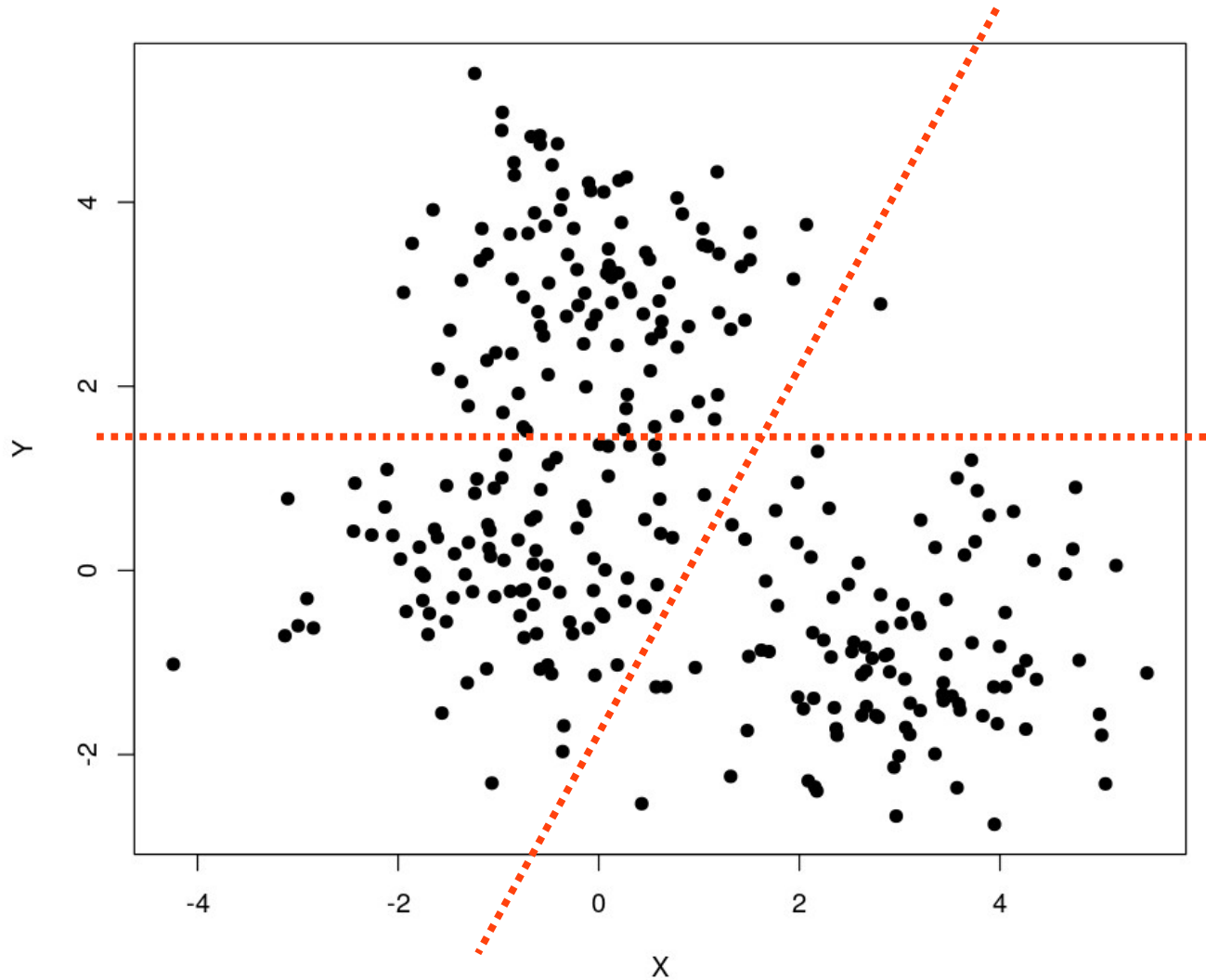
Add sample to cluster by:

- 1) Closest neighbor in cluster (single linkage)
- 2) Farthest neighbor in cluster (complete linkage)
- 3) Median / mean of cluster (UPGMA / UPGMC)
- 4) Least change in variation (Ward)

Disadvantages of hierarchical clustering (no matter type):

Very sensitive to noise in data

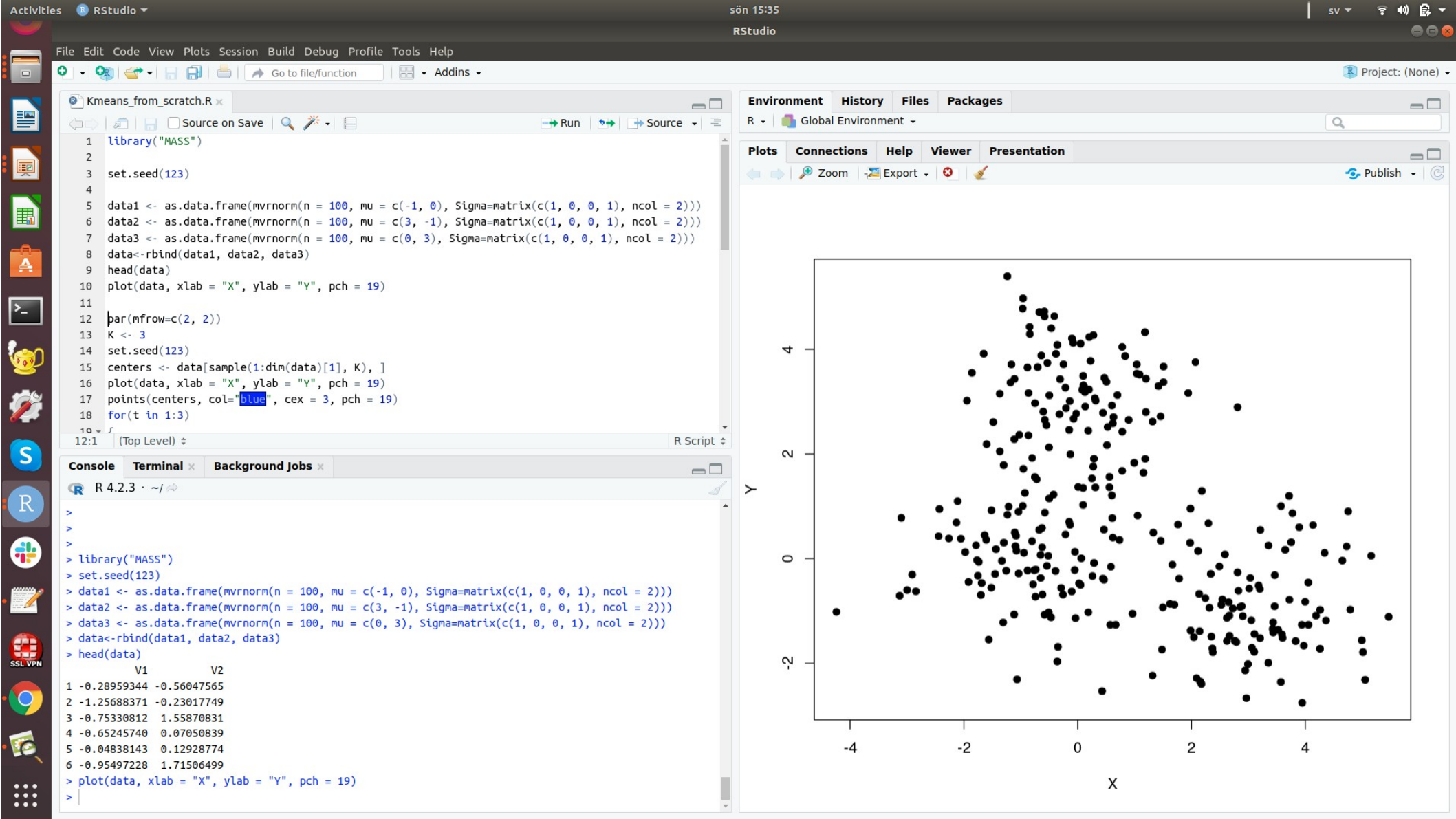
Partitioning clustering

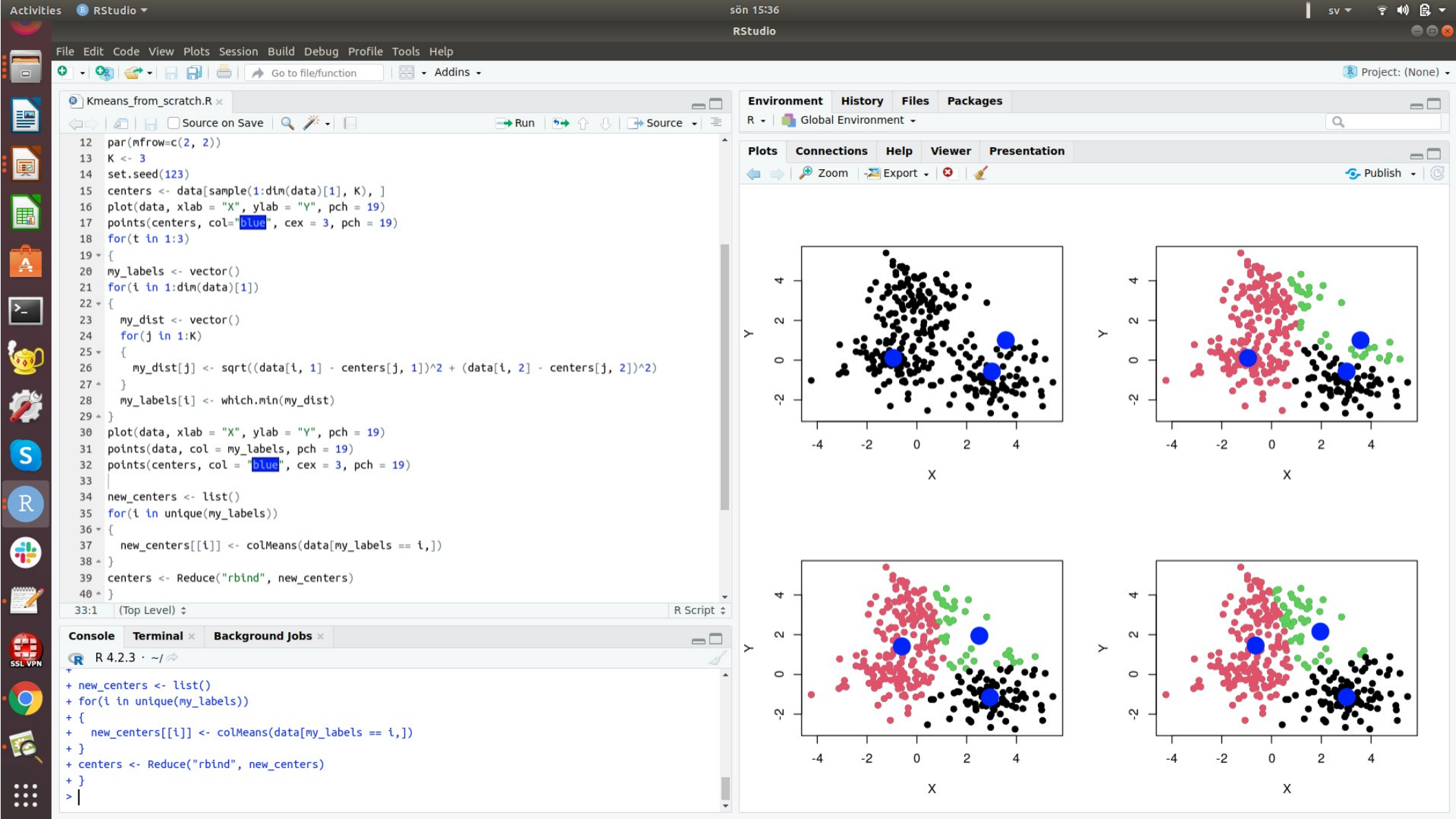


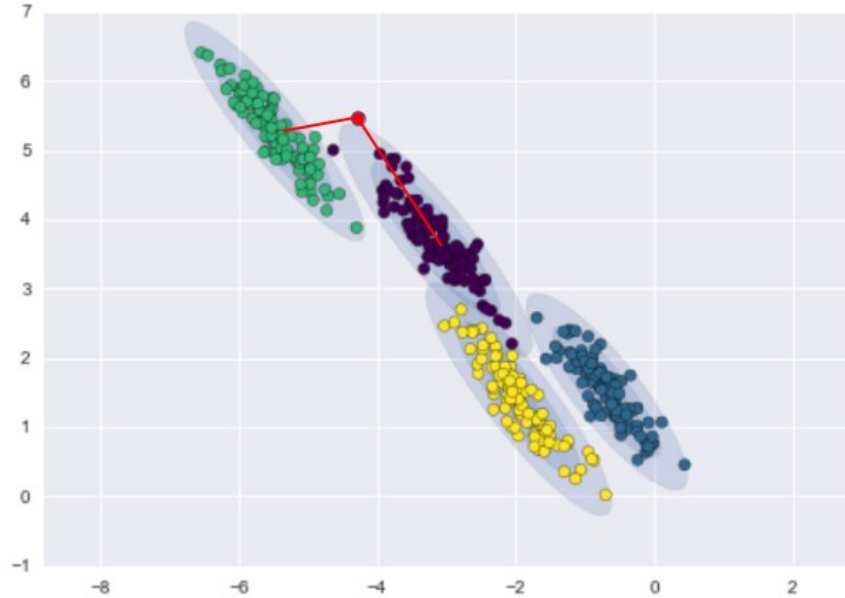
Assign points to
 $K = 2, 3, 4, 5, \dots$
 groups (clusters)

No dendrogram!

K-means, C-means,
 PAM, GMM, ...







GMM is a probabilistic clustering

Expectation-Maximization (EM) algorithm:

$$L(X|\mu_k, \sigma_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(X - \mu_k)^2}{2\sigma_k^2}}$$

Once we computed the likelihood $L(X|\mu, \sigma)$, which we assume follows the Normal distribution, of observing the data given the parameters (means and variances), we can compute the posterior probability of data point assignment to each cluster using the Bayes rule.

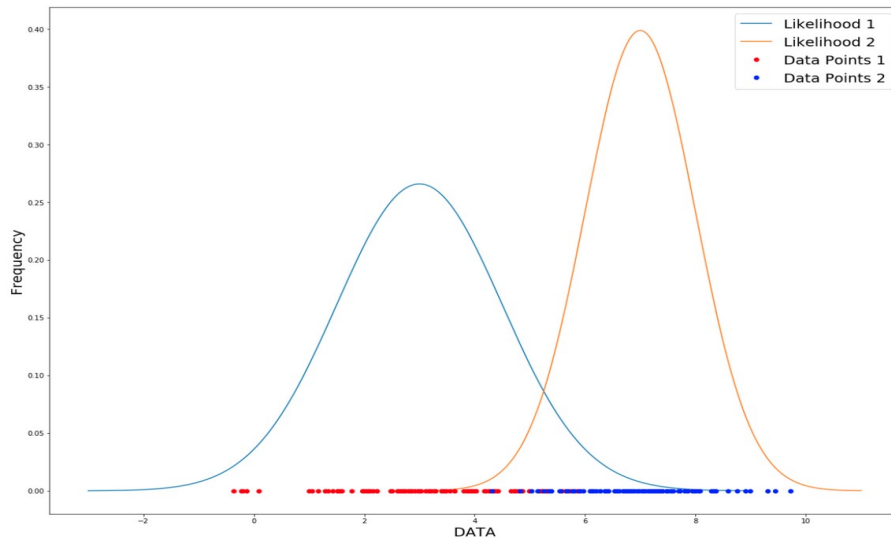
$$p_k(\mu_k, \sigma_k|X) = \frac{L(X|\mu_k, \sigma_k)\phi_k(\mu_k, \sigma_k)}{\sum_k L(X|\mu_k, \sigma_k)\phi_k(\mu_k, \sigma_k)}$$

Next, differentiating the posterior probability with respect to the parameters, means and variances, i.e. by maximizing the posterior probability of the parameters given the data, we can derive an rule for updating the means and variances of the clusters.

$$\mu_k = \frac{\sum_i X_i p_k(\mu_k, \sigma_k|X_i)}{\sum_i p_k(\mu_k, \sigma_k|X_i)}, \quad \sigma_k^2 = \frac{\sum_i (X_i - \mu_k)^2 p_k(\mu_k, \sigma_k|X_i)}{\sum_i p_k(\mu_k, \sigma_k|X_i)}, \quad \phi_k = \frac{1}{N} \sum_i p_k(\mu_k, \sigma_k|X_i)$$

Disadvantages of partitioning clustering (no matter type):

- 1) Assumes spherical / ellipsoidal symmetry of clusters
- 2) Number of clusters has to be defined a-priori



```
import random
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

n = 100
mu1 = 3; sigma1 = 1.5
mu2 = 7; sigma2 = 1

x1 = np.random.normal(mu1, sigma1, n)
x2 = np.random.normal(mu2, sigma2, n)
X = np.array(list(x1) + list(x2))
np.random.shuffle(X)
print("Dataset shape:", X.shape)

x1_approx = np.linspace(mu1 - 4*sigma1, mu1 + 4*sigma1, 10000)
x2_approx = np.linspace(mu2 - 4*sigma2, mu2 + 4*sigma2, 10000)
X_approx = np.array(list(x1_approx) + list(x2_approx))

def pdf(data, mean, sigma):
    return (1 / np.sqrt(2 * np.pi * sigma**2)) * np.exp(- (np.square(data - mean) / (2 * sigma**2)))

plt.figure(figsize = (20,15))
#plt.plot(x1_approx, norm.pdf(x1_approx, mu1, sigma1))
#plt.plot(x2_approx, norm.pdf(x2_approx, mu2, sigma2))
plt.plot(x1_approx, pdf(x1_approx, mu1, sigma1))
plt.plot(x2_approx, pdf(x2_approx, mu2, sigma2))

plt.plot(x1, list(0 for i in range(0, len(list(x1))))), 'ro')
plt.plot(x2, list(0 for i in range(0, len(list(x2))))), 'bo')

plt.ylabel('Frequency', fontsize = 20)
plt.xlabel('DATA', fontsize = 20)
plt.legend(['Likelihood 1', 'Likelihood 2', 'Data Points 1', 'Data Points 2'], loc = 'upper right', fontsize = 20)
plt.show()
```

Coding Expectation-Maximization (EM) algorithm from scratch in Python:

```
k = 2
np.random.seed(123)
means = np.random.choice(X, k)
sigmas = np.random.random_sample(size = k)
priors = np.ones((k)) / k
print(means, sigmas, priors)
```

```
from sklearn import cluster, datasets, mixture
from scipy.stats import multivariate_normal
```

```
N_iter = 100
bins = np.linspace(np.min(X_approx), np.max(X_approx), 10000)
for step in range(N_iter + 1):
    if step % 10 == 0:
```

```
plt.figure(figsize = (20, 15))
plt.ylabel('Frequency', fontsize = 20)
plt.xlabel('DATA', fontsize = 20)
plt.title("Iteration {}".format(step), fontsize = 20)
```

```
plt.plot(x1_approx, pdf(x1_approx, mu1, sigma1))
plt.plot(x2_approx, pdf(x2_approx, mu2, sigma2))
```

```
plt.plot(x1, list(0 for i in range(0, len(list(x1))))), 'ro')
plt.plot(x2, list(0 for i in range(0, len(list(x2))))), 'bo')
```

```
print(means[0], means[1])
print(sigmas[0], sigmas[1])
```

```
plt.plot(bins, pdf(bins, means[0], sigmas[0]), color = 'darkgreen')
plt.plot(bins, pdf(bins, means[1], sigmas[1]), color = 'darkgreen')
```

```
plt.legend(['Likelihood 1', 'Likelihood 2', 'Data Points 1', 'Data Points 2', 'Approximation'],
           loc = 'upper right', fontsize = 20)
plt.show()
```

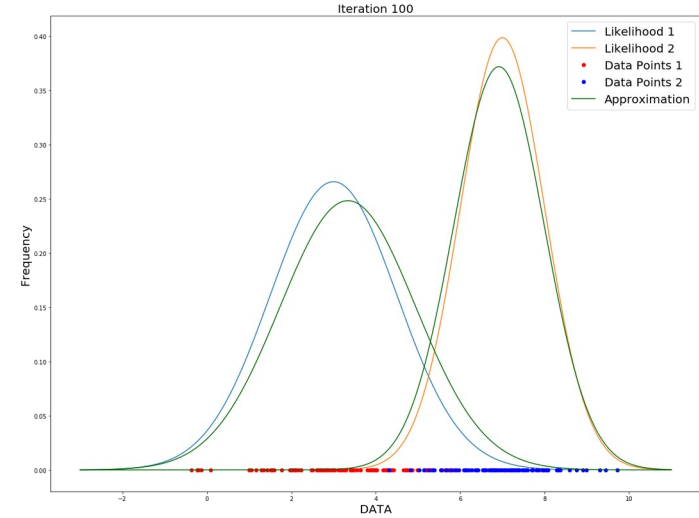
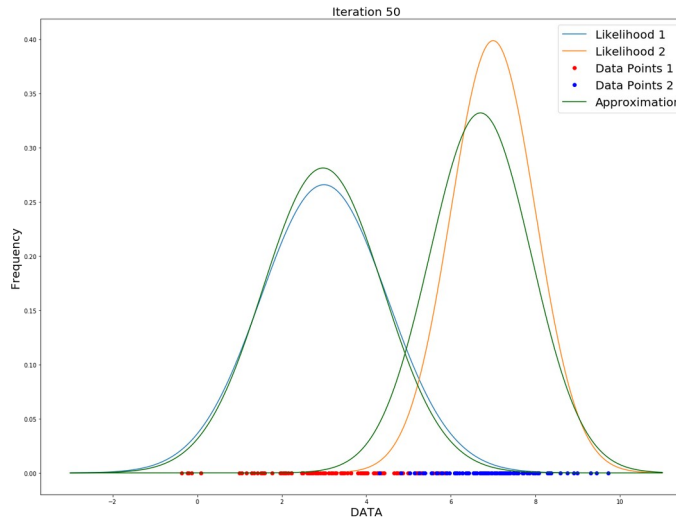
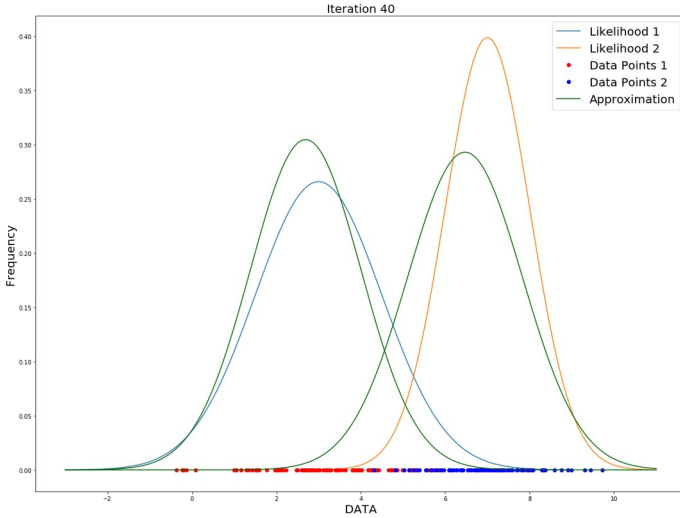
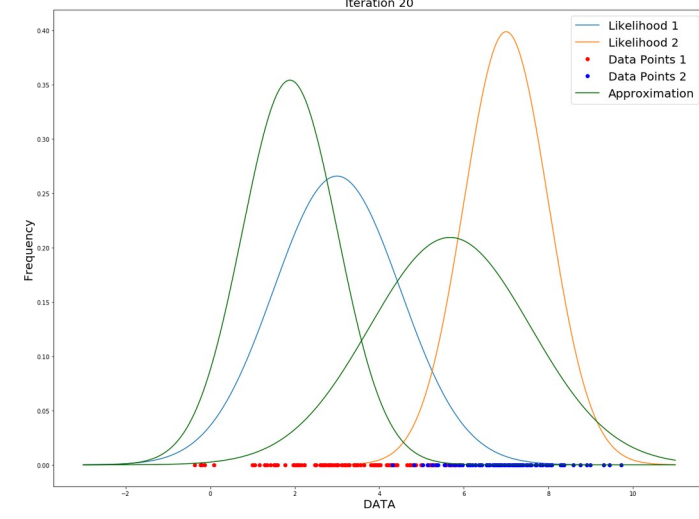
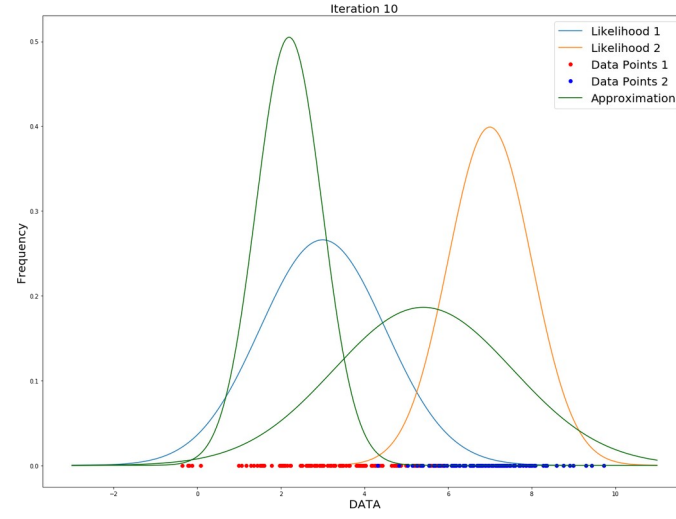
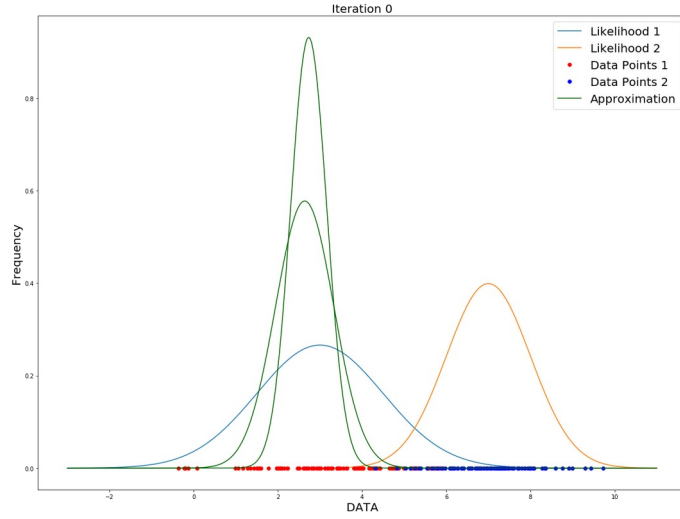
Expectation step

```
likelihood = []
for j in range(k):
    likelihood.append(pdf(X, means[j], sigmas[j]))
```

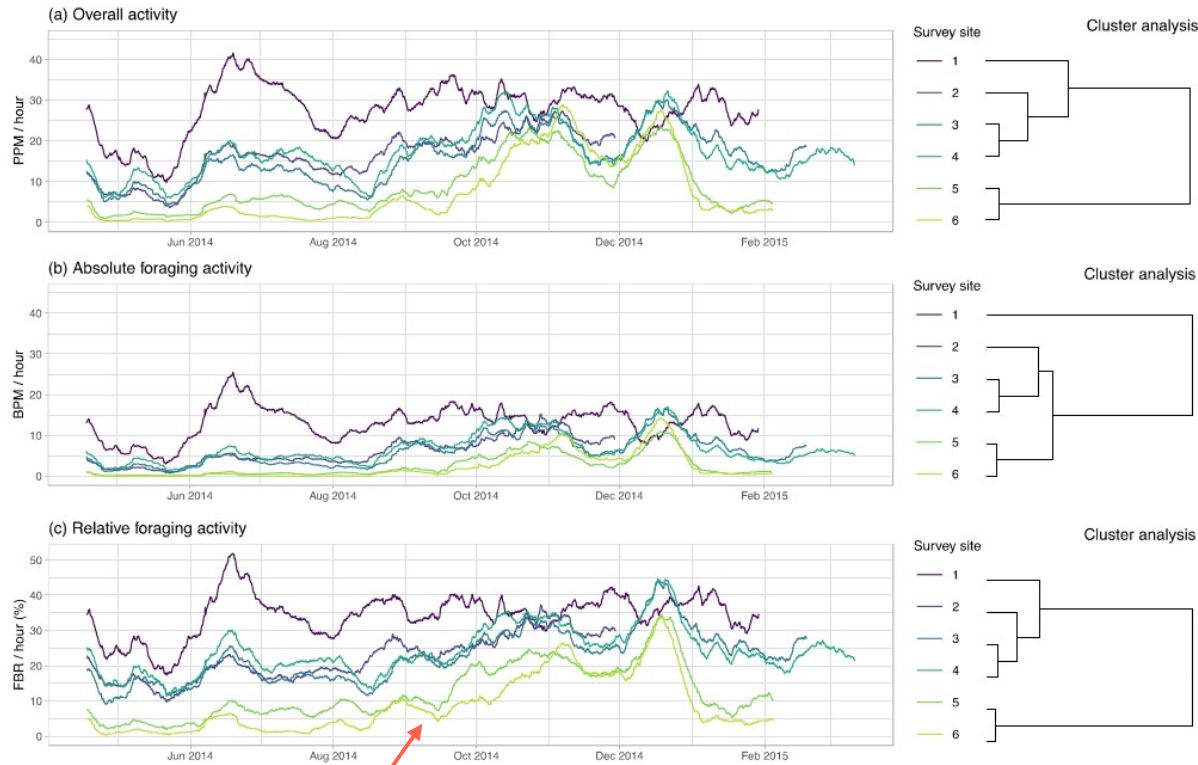
Maximization step

```
posterior = []
for j in range(k):
    posterior.append((likelihood[j] * priors[j]) / (np.sum([likelihood[i] * priors[i] for i in range(k)],
                                                           axis = 0)))

means[j] = np.sum(posterior[j] * X) / np.sum(posterior[j])
sigmas[j] = np.sqrt(np.sum(posterior[j] * np.square(X - means[j])) / np.sum(posterior[j]))
priors[j] = np.mean(priors[j])
```



**Clustering time series data:
examples from epidemiology
and marine ecology**



Foraging activity of
harbour porpoises
is time series data

Vol. 708: 143–161, 2023
<https://doi.org/10.3354/meps14268>

MARINE ECOLOGY PROGRESS SERIES
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Published March 23



Micro-scale spatial preference and temporal cyclicity linked to foraging in harbour porpoises

J. Stedt^{1,*}, M. Wahlberg², J. Carlström³, P. A. Nilsson¹, M. Amundin⁴,
N. Oskolkov^{1,5}, P. Carlsson¹

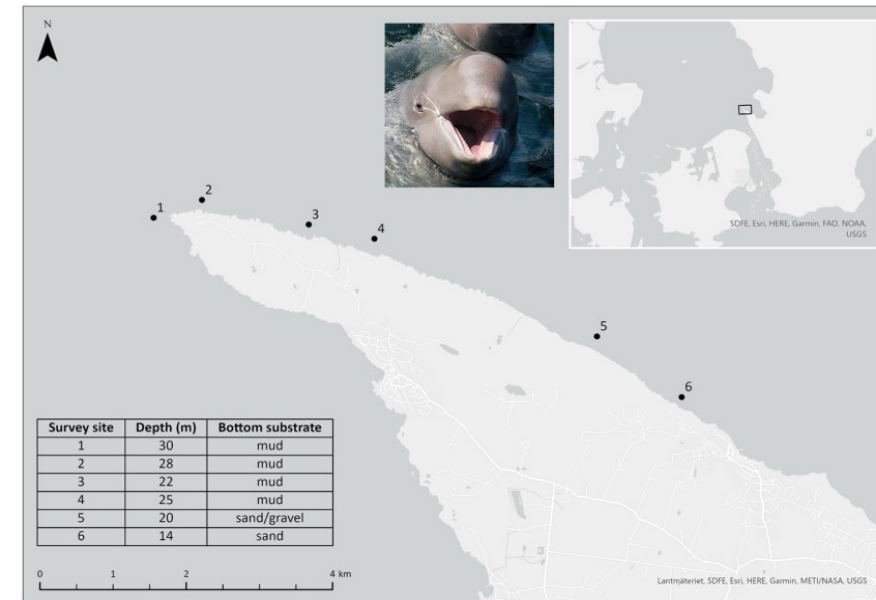
¹Department of Biology, Lund University, 223 62 Lund, Sweden

²Department of Biology, University of Southern Denmark, 5230 Odense, Denmark

³Department of Environmental Research and Monitoring, Swedish Museum of Natural History, 104 05 Stockholm, Sweden

⁴Kolmården Wildlife Park, 618 93 Kolmården, Sweden

⁵National Bioinformatics Infrastructure Sweden (NBIS), Science for Life Laboratory (SciLifeLab), 751 24 Uppsala, Sweden

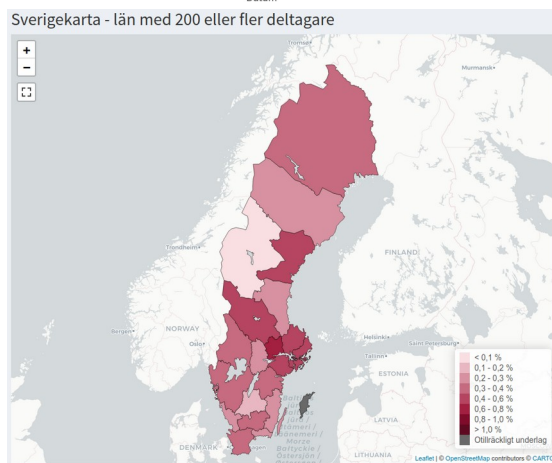
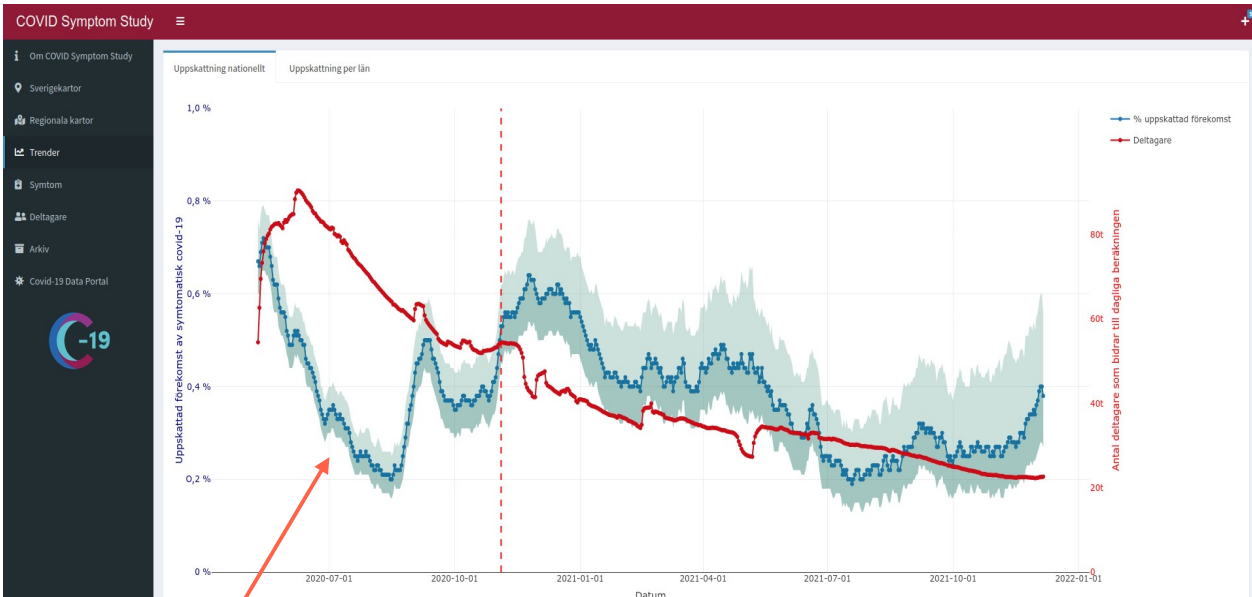


App-based COVID-19 syndromic surveillance and prediction of hospital admissions in COVID Symptom Study Sweden

[Beatrice Kennedy](#), [Hugo Fitipaldi](#), [Ulf Hammar](#), [Marlena Maziarz](#), [Neli Tsereteli](#), [Nikolay Oskolkov](#), [Georgios Varotsis](#), [Camilla A. Franks](#), [Diem Nguyen](#), [Lampros Siliopoulos](#), [Hans-Olov Adami](#), [Jonas Björk](#), [Stefan Engblom](#), [Katja Fall](#), [Anna Grimby-Ekman](#), [Jan-Eric Litton](#), [Mats Martinell](#), [Anna Oudin](#), [Torbjörn Sjöström](#), [Toomas Timpka](#), [Carole H. Sudre](#), [Mark S. Graham](#), [Julien Lavigne du Cadet](#), [Andrew T. Chan](#), ... [Tove Fall](#) + Show authors

[Nature Communications](#) **13**, Article number: 2110 (2022) | [Cite this article](#)

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COVID Symptom Study Sverige - Dashboard

208 541
Deltagare i Sverige



25 070
Aktiva deltagare*



0,38 % (0,27-0,56)
Uppskattad förekomst av
symtomatisk covid-19



17,7 miljoner
Antal rapporter i Sverige



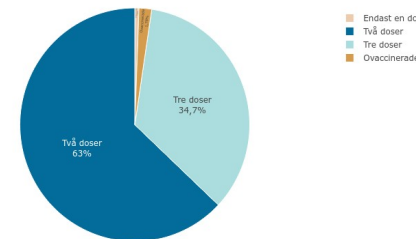
Om Dashboard

Välkommen till COVID Symptom Study Sveriges dashboard med interaktiv grafik. Du navigerar via meny till vänster där du hittar våra senaste nationella och regionala kartor. Använd datumreglaget för att välja det datum du vill visa för kartorna. Vi arbetar hela tiden med att utöka funktionaliteten och innehåll på denna sida så vi ber om ert tålamod under tiden som sidan utvecklas. För frågor och feedback - [mejla oss](#)

Information

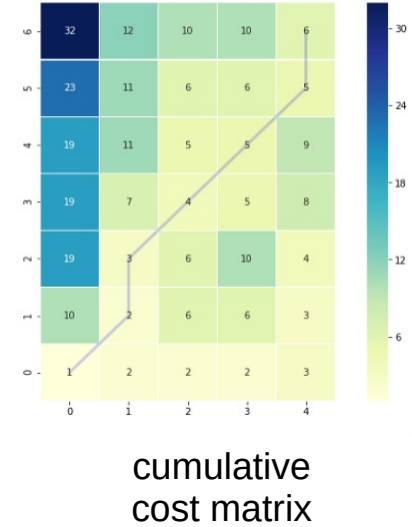
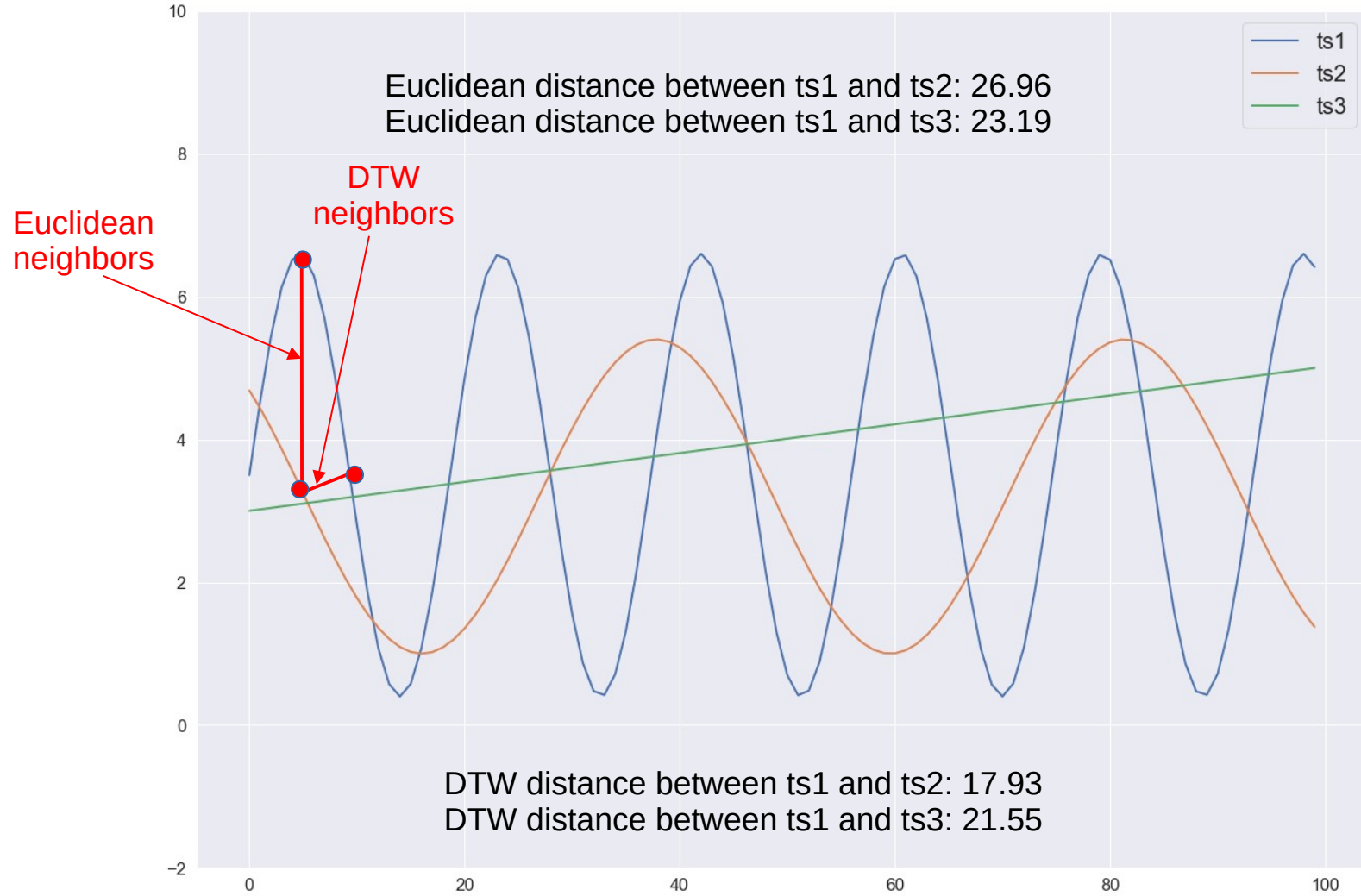
Vi har nu analyserat hur många av våra aktiva deltagare som är vaccinerade. Då hela 98 % av våra aktiva studiedeltagare nu är vaccinerade med två doser vaccin, och 35 % med tre doser, så är våra prediktioner för förekomsten av symtomatisk covid-19 främst överförbara på den vuxna vaccinerade befolkningen i Sverige.

% vaccinerade bland aktiva deltagare*



*aktiva deltagare är de som bidragit med data till studien minst en gång under de senaste sju dagarna

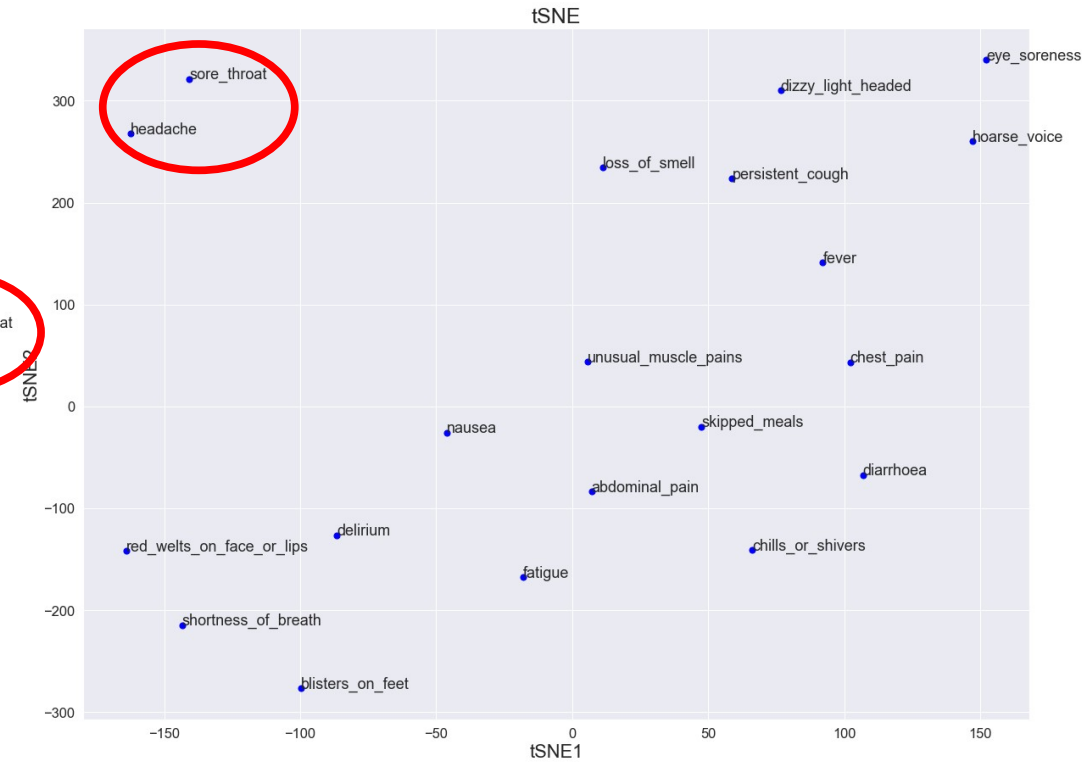
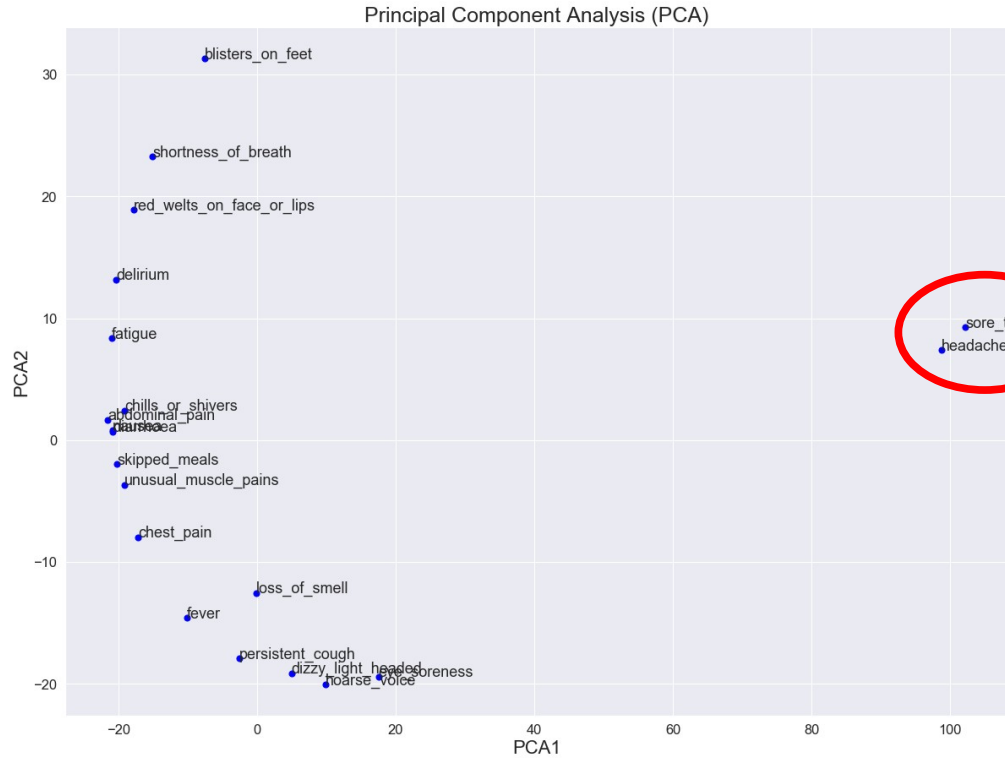
Covid prevalence
in Sweden in time
is time series data



```
my_symptoms = list(d.columns)
my_distance = list()
for i in my_symptoms:
    i_symptom = d.T.loc[i]
    for j in my_symptoms:
        j_symptom = d.T.loc[j]
        distance, _ = fastdtw(i_symptom, j_symptom, dist = euclidean)
        if i == j:
            distance = 0
        my_distance.append(distance)
my_distance_df = np.array(my_distance).reshape(len(my_symptoms), len(my_symptoms))
symptom_dist = pd.DataFrame(my_distance_df, columns = my_symptoms, index = my_symptoms)
symptom_dist.iloc[0:10, 0:10]
```

Out[15]:

	persistent_cough	fatigue	delirium	shortness_of_breath	fever	headache	nausea	eye_soreness	chills_or_shivers	dizzy_light
persistent_cough	0.000000	9.097296	10.292449	13.267313	1.346679	24.329388	7.004967	4.567481	6.535234	
fatigue	9.105655	0.000000	1.463343	4.064644	6.639465	33.457821	2.841472	14.065845	2.901188	
delirium	10.300808	1.464591	0.000000	2.808816	7.975508	34.652974	3.061681	15.424051	4.122948	
shortness_of_breath	13.267313	4.043686	2.794971	0.000000	11.532179	37.596701	6.238593	18.293569	5.919232	
fever	1.325948	6.471745	7.734869	11.532179	0.000000	26.064522	4.019190	6.792987	4.003027	
headache	24.329388	33.457821	34.652974	37.596701	26.064522	0.000000	31.361039	19.318277	32.329450	
nausea	7.000836	2.758465	2.942926	6.240409	4.005763	31.361039	0.000000	12.042762	3.083702	
eye_soreness	4.490836	14.110260	15.431009	18.284482	6.745880	19.318277	12.042762	0.000000	12.879221	
chills_or_shivers	6.643880	2.886984	4.117598	5.889634	3.930427	32.329450	3.044511	12.799639	0.000000	
dizzy_light_headed	1.915427	10.891548	12.234134	15.061940	3.526857	22.528484	8.832555	2.154606	9.852064	



Clustering covid19 symptom trajectories by Dynamic Time Warping

